Pascal's triangle

1

a $1, 5, 10, 10, 5, 1$

b 32

c ${}_5C_0, {}_5C_1, {}_5C_2, {}_5C_3, {}_5C_4, {}_5C_5$

2 $1, 8, 28, 56, 70, 56, 28, 8, 1$

3 7th

4

a 7

b 9

5

a 5

b 4

6

a $(1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$

b $(x+m)^4 = 1x^4m^0 + 4x^3m^1 + 6x^2m^2 + 4x^1m^3 + 1x^0m^4$

c $(5+b)^5 = 1 \times 5^5b^0 + 5 \times 5^4b^1 + 10 \times 5^3b^2 + 10 \times 5^2b^3 + 5 \times 5^1b^4 + 1 \times 5^0b^5$

d $(ka-b)^6 = 1(ka)^4(-b)^0 + 6(ka)^3(-b)^1 + 15(ka)^2(-b)^2 + 20(ka)^1(-b)^3 + 15(ka)^0(-b)^4 + 6(ka)^0(-b)^5 + 1(ka)^0(-b)^6$

7

a $(p+q)^5 = p^5 + 5p^4q + 10p^3q^2 + 10p^2q^3 + 5pq^4 + q^5$

b $(c-d)^8 = c^8 - 8c^7d + 28c^6d^2 - 56c^5d^3 + 70c^4d^4 - 56c^3d^5 + 28c^2d^6 - 8cd^7 + d^8$

c $(y+3)^4 = y^4 + 12y^3 + 54y^2 + 108y + 81$

d $(u+4v)^6 = 4096v^6 + 6144uv^5 + 3840u^2v^4 + 1280u^3v^3 + 240u^4v^2 + 24u^5v + u^6$

e $\left(1 + \frac{4}{y}\right)^3 = 1 + \frac{12}{y} + \frac{48}{y^2} + \frac{64}{y^3}$

8

a 6

b 14

c 10

d 1

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Binomial theorem

9 Negative

10 $(4p+3q)^3 = {}_3C_0\,(4p)^3\,(3q)^0 + {}_3C_1\,(4p)^2\,(3q)^1 + {}_3C_2\,(4p)^1\,(3q)^2 + {}_3C_3\,(4p)^0\,(3q)^3$

11

a $256x^4$ **b** $81y^4$

12 $55u^2v^9, 55u^9v^2$

13

a $560b^3$ **b** $126c^5d^4$ **c** $135u^2v^4$ **d** $\frac{55}{128}u^3$

e $\frac{280}{3y^2}$ **f** $2288x^3y^{20}$ **g** $-3876x^4y^{75}$ **h** $1\,451\,520x^8y^{12}$

i $216x^{-6}y^{-4}$ **j** 1792

14

a $1+4x+6x^2+4x^3+x^4$

b $32+80b+80b^2+40b^3+10b^4+b^5$

c $p^5+5p^4q+10p^3q^2+10p^2q^3+5pq^4+q^5$

d $c^8-8c^7d+28c^6d^2-56c^5d^3+70c^4d^4-56c^3d^5+28c^2d^6-8cd^7+d^8$

e $u^6+24u^5v+240u^4v^2+1280u^3v^3+3840u^2v^4+6144uv^5+4096v^6$

f $y^4+12y^3+54y^2+108y+81$

g $(ka)^6-6\,(ka)^5\,b+15\,(ka)^4\,b^2-20\,(ka)^3\,b^3+15\,(ka)^2\,b^4-6kab^5+b^6$

h $y^3-15y^2+75y-125$

i $y^3-y^2+\frac{y}{3}-\frac{1}{27}$

j $8+6y+\frac{3y^2}{2}+\frac{y^3}{8}$

k $27y^3+54ry^2+36r^2y+8r^3$

l $a^3-3a+\frac{3}{a}-\frac{1}{a^3}$

m $u^6+9u^4v^2+27u^2v^4+27v^6$

n $1+\frac{12}{y}+\frac{48}{y^2}+\frac{64}{y^3}$

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15 $45x^2y^8, 10xy^9, y^{10}$

16

a $n = 10$ **b** $45m^2k^8$

17 $n = 16$

18 $\frac{21}{2}$

19 ${}_9C_3$

20

a $405x^4$ **b** $120c^3d^7$ **c** $165u^3v^8$ **d** $3003u^5v^{10}$

21 $10\,777\,536$

22 9

23 $p = 2$

24 ${}_4C_3$

25

a ${}_5C_2$ **b** 10 **c** 4th term

26 ${}_8C_k \times 3^{8-k}\,(-x)^k$

27

a 0.943 **b** 1.037 **c** 0.890

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